Journal of Psychiatric Research 189 (2025) 71–75

Contents lists available at ScienceDirect

Journal of Psychiatric Research

journal homepage: www.elsevier.com/locate/jpsychires

The association between aggression and suicidal behaviors in veterans at risk for suicide

Alison Krauss[a,b,*], Michael S. McCloskey[c], A. Solomon Kurz[a,b], Mark A. Ilgen[d,e], Suzannah K. Creech[a,b,f], Marianne Goodman[a,g,h]

[a] VA VISN 17 Center of Excellence for Research on Returning War Veterans, Waco, TX, USA
[b] Central Texas Veterans Health Care System, Temple, TX, USA
[c] Department of Psychology, Temple University, Philadelphia, PA, USA
[d] VA Center for Clinical Management Research (CCMR), Ann Arbor, MI, USA
[e] Department of Psychiatry, University of Michigan, Ann Arbor, MI, USA
[f] Department of Psychiatry and Behavioral Sciences, Dell Medical School of the University of Texas at Austin, Austin, TX, USA
[g] VA VISN 2 Mental Illness Research, Education and Clinical Center, James. J. Peters VA Medical Center, Bronx, NY, USA
[h] Department of Psychiatry, Icahn School of Medicine, Mount Sinai Hospital, New York, NY, USA

ARTICLE INFO

Keywords:
Suicide behavior
Aggression
Veterans

ABSTRACT

Suicide is a leading cause of death among United States veterans. Although aggression is common among veterans and represents a strong predictor of suicidal behaviors in civilians, it is seldom studied among veterans. The scant existing research on aggression and suicide in veterans is limited by its restricted examination of the full range of suicidal behavior, oversimplification of suicide attempt history (no attempt vs. attempt), and its focus on physical aggression rather than examining both physical and verbal aggression as unique correlates of suicide. The current study addresses these gaps in a sample of veterans at high-risk for suicide. Participants ($N = 207$) were recruited as part of a larger clinical trial examining a suicide intervention; the current study uses the baseline data. Veterans completed a self-report measure of physical and verbal aggression and a clinician-administered interview of suicidal behaviors. Logistic regression models indicated that aggression was unrelated to the presence of suicide attempts or behaviors in this high-risk sample. However, count models suggested that physical and verbal aggression were related to a greater number of suicide attempts, and physical aggression was related to a greater number of suicidal behaviors. The current findings align with civilian research suggesting aggression is a risk factor for frequent suicide attempts. Further research is needed to better understand this association, particularly in determining whether aggression plays a causal role in suicide behaviors in veterans.

1. Background

Suicide is a leading cause of death among veterans (U.S. Department of Veterans Affairs, 2023), and suicide prevention is the top clinical priority of the Department of Veterans Affairs (VA; U.S. Department of Veterans Affairs, 2019). Yet from 2020 to 2021, the U.S. veteran suicide rate increased by 11.6 %, more than doubling the suicide rate increase of 4.5 % among non-veteran U.S. adults (U.S. Department of Veterans Affairs, 2023). Aggression represents a key predictor of suicide attempts among civilians (Gvion and Levi-Belz, 2018) and active-duty military personnel (Schafer et al., 2022). Aggression is also common among veterans (Elbogen et al., 2012), but it is seldom studied as a predictor of veteran suicide. The current study extends existing research by examining the association between aggression and suicide attempts in a sample of veterans at high-risk for suicide.

1.1. Aggression and suicidal behaviors

A wealth of research among civilians identifies aggression as a predictor of suicidal behaviors (Gvion, 2018; McCloskey and Ammerman, 2018). In fact, the association between aggression and suicide is so robust among civilians that aggression is recognized as a primary attribute of a common suicide phenotype: one characterized by high levels of aggression and frequent suicide attempts (Oquendo et al., 2021; Stanley et al., 2019). Related constructs such as anger (a common precursor to aggression) are also associated with suicide attempts (Dillon

* Corresponding author. VISN 17 Center of Excellence for Research on Returning War Veterans, 4800 Memorial Drive (151C), Waco, TX, 76711, USA.
E-mail address: alison.krauss@va.gov (A. Krauss).

https://doi.org/10.1016/j.jpsychires.2025.06.001
Received 6 February 2025; Received in revised form 28 May 2025; Accepted 2 June 2025
Available online 3 June 2025

et al., 2020). Several theories offer explanations for the association between aggression and suicide, including shared genetic risk (Mann et al., 2009), common underpinnings in psychopathology (Gvion and Apter, 2011), and a causal link through the deleterious consequences of aggression (i.e., damage to interpersonal relationships, loss of employment, incarceration; Conner et al., 2003).

Aggression may also be an important predictor of suicide attempts in military samples. In a recent meta-analysis of longitudinal suicide risk factors among military personnel, aggression emerged as one of the five strongest predictors of dying by suicide among active-duty personnel (Schafer et al., 2022). Notably, the authors did not identify any studies using veteran samples. This stands in stark contrast to the high prevalence rates of aggression among veterans, with 33% of post-9/11 veterans engaging in physical aggression (e.g., kicking, slapping, using fists) and 11% engaging in severe violence (e.g., using a weapon against someone) in the past year (Elbogen et al., 2012).

Only a handful of cross-sectional studies have documented a positive association between aggression and suicide attempts among veterans (McGlade et al., 2021; Watkins et al., 2017), with one exception (Varker et al., 2022). Additionally, no studies on aggression and suicide among veterans have considered the range of suicidal behaviors, such as aborted or interrupted attempts. Aborted and interrupted attempts occur when an individual initiates an action to end their life, but is stopped by themselves or someone else (Posner et al., 2011). Individuals who report aborted and interrupted attempts are at risk for suicide attempts (Chu et al., 2023), and there are few differences in risk factors for interrupted, aborted, and suicide attempts (Burke et al., 2016). Though it seems likely aggression is positively associated with the range of suicide behaviors, this suggestion has not been tested.

Furthermore, prior research on the association between aggression and suicide attempts among veterans measures suicide attempts as a binary variable rather than assessing the number of suicide attempts. Given that aggression is associated with making frequent suicide attempts among civilians (Oquendo et al., 2021; Stanley et al., 2019), it seems likely that aggression is associated with both the presence and number of suicidal behaviors among veterans. Most of the existing research with veterans has explored the association between physical aggression and suicide. Though often co-occurring, physical aggression and verbal aggression are distinct (Dewi and Kyranides, 2022; Hillbrand, 2001). Thus, whether different forms of aggression are uniquely associated with suicidal behaviors among veterans remains unknown. Finally, although one study has examined the co-occurrence of aggression and suicide attempts in a clinical sample (Watkins et al., 2017), no studies have done so in samples of veterans at high-risk for suicide. Given that veterans at high-risk for suicide are a subpopulation of clinical interest in VA, identifying aggression as a correlate of suicide behaviors will aid in better characterizing this group.

1.2. Current study

The current study extends our understanding of the association between aggression and suicidal behaviors among veterans by: examining this association across both suicide attempts and total suicidal behaviors (including suicide attempts, aborted attempts, and interrupted attempts), testing this association with the presence and number of suicidal behaviors, exploring the independent associations between verbal and physical aggression with suicidal behaviors, and using a sample of veterans at high-risk for suicide. We hypothesized that: 1) both physical aggression and verbal aggression would be independently, positively associated with the presence of suicide attempts and total suicidal behaviors, and 2) physical aggression and verbal aggression would be independently, positively associated with a greater number of suicide attempts and total suicidal behaviors.

2. Method

2.1. Participants

Participants were recruited as part of a parent study evaluating a suicide safety planning intervention; the current study uses baseline data from that trial. Veterans were recruited across four VA sites: three located in the northeast and one in the southwest. Study staff recruited participants through local VA electronic medical records and clinician referrals. To be eligible for the study, veterans must have been deemed at high-risk for suicidal behaviors, defined as meeting one of the following criteria: 1) discharged from an inpatient unit for suicidal ideation or attempts within the past month, 2) placed on a VA high-risk suicide flag, 3) scored a 3 or higher (out of 5) on the suicidal ideation subscale of the Columbia – Suicide Severity Rating Scale (C-SSRS) at baseline, or 4) made a suicide attempt in the past year. For further details, see clinicaltrials.gov Identifier NCT03653637. A total of 207 participants completed the baseline assessment and were included in the below analyses.

2.2. Procedures

The institutional review board at each of the participating VA sites approved all study procedures. Recruitment and study procedures occurred between 2018 and 2023. All study procedures occurred in person at the local VA site prior to 2020; study procedures were moved to VA-approved virtual platforms in March 2020. Veterans provided informed consent prior to their baseline appointment. Participants were compensated $75 for completion of their baseline assessment.

2.3. Measures

2.3.1. Aggression

Veterans reported on their trait aggression using the Physical Aggression and Verbal Aggression subscales of the Buss-Perry Aggression Questionnaire (BPAQ; Buss and Perry, 1992). Participants were asked to rate how characteristic each item was of them on a 5-point scale ranging from *not at all like me* (1) to *very much like me* (5). The Physical Aggression subscale includes 9 items (e.g., "Given enough provocation, I may hit another person"), and the Verbal Aggression subscale includes 5 items (e.g., "When people annoy me, I may tell them what I think of them"). Subscale scores are summed with higher scores indicating greater aggression; reliability in the current sample was $\alpha = .83$ (95 % CI [0.80, 0.87]) for physical aggression and $\alpha = .73$ (95 % CI[0.68, 0.79]) for verbal aggression.

2.3.2. Suicidal behaviors

Veterans indicated their past year suicidal behaviors on the C-SSRS (Posner et al., 2011), a clinician administered interview of suicide. Participants indicated the number of suicide attempts, aborted attempts, and interrupted attempts they had made in the past year. We created four scores from C-SSRS interviews. First, we created a presence of suicide attempts score (0 = no suicide attempts in the past year, 1 = one or more suicide attempts in the past year). Second, we created a number of suicide attempts score representing the number of reported suicide attempts. Third, we created a presence of total suicidal behaviors score by combining reports across suicide attempts, aborted attempts, and interrupted attempts (0 = no suicidal behaviors in the past year, 1 = one or more suicidal behaviors). Finally, consistent with previous research using the C-SSRS (Miller et al., 2017; Posner et al., 2011), we summed the reported number of suicide attempts, aborted attempts, and interrupted attempts to create a total suicidal behaviors score.

2.4. Missing data and data analysis plan

About 3% of participants had missing data on the BPAQ, 5% had missing data on suicide attempts, and 6% had missing data on total suicide behaviors. We used fully conditional specification multiple imputation via the *mice* package in R (van Buuren and Groothuis-Oudshoorn, 2011). We imputed data at the item level using predictive mean matching and computed sum scores through passive imputation. Per Enders (2022), we included all main study variables in the imputation of each other, as well as salient auxiliary variables. We imputed 100 datasets and used Rubin's rules for pooling estimates after multiple imputation (Enders, 2022).

In all main analyses, we used standardized scores (*z*-scores) for physical and verbal aggression. To examine hypothesis 1, we fit two logistic regressions: one with presence of suicide attempts as the outcome and one with presence of total suicidal behaviors as the outcome. To examine hypothesis 2, we conducted a similar approach but fit zero-inflated negative binomial models. Negative binomial models are used for count data when the variance of the dependent variable is significantly larger than its mean, and zero-inflated models are used when the number of "0" scores likely exceeds the number of zeros expected in a distribution (Yau et al., 2003). A simulation-based power analysis indicated that with a sample size of 207, we had sufficient power (>0.80) to detect a small effect in logistic regression models (difference of 10 % in probability) and negative binomial models (difference of 0.5 in count).

3. Results

3.1. Sample characteristics descriptive statistics

Table 1 presents demographic characteristics and Supplemental Table 1 reports military characteristics of the sample. Approximately 52% of veterans reported at least one suicidal behavior and 33% reported a suicide attempt in the past year. The number of total suicidal behaviors ranged from 0 to 15 and the number of suicide attempts ranged from 0 to 5. Supplemental Table 2 presents descriptive statistics and correlations between main study variables. Physical and verbal aggression were positively correlated with each other (r = 0.65) and with the number of suicide attempts (r = 0.16–0.19), but not other suicide related variables (r = 0.07–0.11).

3.2. Association between aggression and presence of suicide attempts and behaviors

Neither physical nor verbal aggression were associated with the presence of suicide attempts when examined in separate models (Table 2). When including both subscales in the same model, neither were associated with the presence of suicide attempts. Similarly, physical and verbal aggression were unrelated to the presence of total suicidal behaviors in separate models, and neither were associated with total suicidal behaviors when entered as concurrent predictors.

3.3. Association between aggression and number of suicide attempts and behaviors

Physical aggression and verbal aggression were positively associated with the number of suicide attempts when entered in separate models (Table 2). When both were entered into the same model, neither were related to the number of suicide attempts. Physical aggression was also positively associated with the number of total suicidal behaviors when entered as the only predictor; however verbal aggression alone was not associated with the number of total suicidal behaviors. When physical aggression and verbal aggression were entered into the same model, neither were related to the number of total suicidal behaviors.

3.4. Exploratory analysis

To explore whether our non-significant findings when physical and verbal aggression were entered into models together were due to shared variance between physical and verbal aggression, we fit our zero-inflated negative binomial models using a composite score of physical and verbal aggression (computed by summing the physical aggression and verbal aggression items on the BPAQ). The composite aggression score was positively associated with number of attempts (B = 0.30, SE = 0.12, 95 % CI[0.06, 0.53], p = .01) but not with number of suicidal behaviors (B = 0.20, SE = 0.11, 95 % CI[-0.01, 0.41], p = .06).

Given that suicide behaviors seemed to differ by some demographic and military characteristics (e.g., race, ethnicity, employment status, and highest military rank; see Table 1 and Supplemental Table 1), we

Table 1
Demographic characteristics by report of suicidal behaviors.

Demographic characteristic	*No suicidal behaviors (n = 95)*	*1+ suicidal behaviors (n = 101)*	*Total sample (n = 207)*
Age (*M* and *SD*)	46.22 (14.37)	45.23 (13.57)	46.13 (13.88)
Gender			
Man	81 (86 %)	82 (81 %)	170 (82 %)
Woman	11 (12 %)	16 (16 %)	28 (14 %)
Not reported	3 (1 %)	3 (1 %)	9 (4 %)
Ethnicity			
Hispanic/Latinx	25 (26 %)	42 (42 %)	69 (33 %)
Non-Hispanic/Latinx	63 (66 %)	55 (55 %)	124 (60 %)
Not reported	7 (3 %)	4 (2 %)	14 (7 %)
Race			
White	43 (45 %)	35 (35 %)	80 (39 %)
Black or African American	33 (35 %)	32 (32 %)	71 (34 %)
Asian American	3 (3 %)	1 (1 %)	4 (2.0 %)
Native American or Alaska Native	0 (0 %)	1 (1 %)	1 (1 %)
Multiracial	11 (12 %)	14 (14 %)	26 (13 %)
Other	5 (5 %)	14 (14 %)	19 (9 %)
Not reported	0 (0 %)	4 (2 %)	6 (3 %)
Education			
Some high school	2 (2 %)	1 (1 %)	3 (1 %)
High school diploma/ GED	30 (32 %)	22 (22 %)	54 (26 %)
Some college or 2-year degree	44 (46 %)	57 (56 %)	104 (50 %)
Bachelors Degree	10 (11 %)	17 (17 %)	30 (15 %)
Masters degree	5 (5 %)	0 (0 %)	5 (2 %)
Not reported	4 (2 %)	4 (2 %)	11 (5 %)
Employment			
Full-time employment	14 (15 %)	21 (21 %)	36 (17 %)
Part-time employment	2 (2 %)	6 (6 %)	8 (4 %)
Self-employed	2 (2 %)	3 (3 %)	5 (2 %)
Student	6 (6 %)	5 (5 %)	12 (6 %)
Retired	33 (35 %)	18 (18 %)	54 (26 %)
Unemployed	33 (35 %)	43 (43 %)	79 (38 %)
Not reported	5 (2 %)	5 (2 %)	13 (6 %)
Physical aggression (*M* and *SD*)	16.35 (9.37)	18.23 (8.60)	17.50 (8.96)
Verbal aggression (*M* and *SD*)	9.63 (4.91)	10.31 (4.76)	10.05 (4.85)
Suicide attempts – presence	0 (0 %)	65 (64 %)	65 (31 %)
Suicidal behaviors – presence	0 (0 %)	101 (100 %)	101 (49 %)
Suicide attempts – number (*M* and *SD*)	0 (0)	0.88 (0.90)	0.45 (0.78)
Suicidal behaviors – number (*M* and *SD*)	0 (0)	1.82 (2.18)	0.93 (1.80)

Note. The amount of missing data differed across demographic variables and suicidal behaviors. Therefore, numbers within the same row or column may not add to the subsample size or total sample size.

Table 2
Regression models predicting the presence of suicide attempts and presence of total suicidal behaviors.

Variable	Logistic Regression Models				Negative Binomial Models			
	Suicide Attempts		Suicidal Behaviors		Suicide Attempts		Suicidal Behaviors	
	B (SE)	*Effect Size*[a]	*B (SE)*	*Effect Size*[a]	*B (SE)*	*Effect Size*[b]	*B (SE)*	*Effect Size*[b]
Intercept	−0.71 (0.15)** [-1.00, −0.41]	–	0.08 (0.14) [-0.21, 0.37]	–	−0.82 (0.12)** [-1.06, −0.58]	–	−0.08 (0.11) [-0.29, 0.14]	–
Physical aggression	0.15 (0.15) [-0.15, 0.46]	0.03–0.04	0.20 (0.15) [-0.09, 0.49]	0.05–0.05	0.26 (0.12)* [0.03, 0.50]	0.12–0.19	0.22 (0.11)* [0.01, 0.43]	0.20–0.22
Intercept	−0.71 (0.15)** [-1.01, −0.41]	–	0.08 (0.14) [-0.21, 0.37]	–	−0.84 (0.12)* [-1.08, −0.59]	–	−0.06 (0.11) [-0.28, 0.16]	–
Verbal aggression	0.20 (0.16) [-0.11, 0.50]	0.04–0.05	0.12 (0.15) [-0.17, 0.40]	0.03–0.03	0.31 (0.12)* [0.07, 0.55]	0.10–0.21	0.14 (0.10) [-0.06, 0.34]	0.12–0.14
Intercept	−0.71 (0.15)** [-1.01, −0.41]	–	0.08 (0.15) [-0.21, 0.37]	–	−0.87 (0.12)* [-1.08, −0.59]	–	−0.08 (0.11) [-0.30, 0.14]	–
Physical aggression	0.04 (0.20) [-0.36, 0.45]	0.01–0.01	0.21 (0.19) [-0.17, 0.59]	0.05–0.05	0.09 (0.16) [-0.23, 0.41]	0.01–0.02	0.22 (0.15) [-0.08, 0.52]	0.18–0.22
Verbal aggression	0.17 (0.21) [-0.24, 0.57]	0.04–0.04	−0.02 (0.19) [-0.40, 0.36]	0.00–0.00	0.25 (0.16) [-0.08, 0.58]	0.10–0.10	0.01 (0.14) [-0.29, 0.28]	0.00–0.00

*$p < .05$, **$p < .01$.

[a] Effect size represents the change in the predicted proportion of participants endorsing suicide attempts or suicide behaviors as the predictor increases from 1 standard deviation (SD) below the mean to 1 SD above the mean. For instance, an effect size of 0.03–0.04 suggests that the proportion of participants endorsing a suicide attempt ranges from 30 % to 40 % of participants as the predictor increases from −1 SD to +1 SD.

[b] Effect size represents the change in the predicted number of reported suicide attempts or suicide behaviors as the predictor increases from −1 SD to +1 SD. For instance, an effect size of 0.12–0.19 suggests that the predictor number of reported suicide attempts ranges from 0.12 attempts to 0.19 attempts as the predictor increases from −1 SD to +1 SD.

reran all models while including these as covariates. The pattern of results replicated our main analysis (see Supplemental Table 3).

4. Discussion

The current study broadens the small body of research examining the association between aggression and suicidal behaviors among veterans. Contrary to our first hypothesis, physical and verbal aggression were not meaningfully associated with the presence of suicide attempts or suicidal behaviors among high-risk veterans. In support of our second hypothesis, physical and verbal aggression were associated with the number of suicide attempts when entered in separate models, and physical aggression was associated with the number of suicidal behaviors. However, these effects were quite small. Taken together, these findings suggest that aggression is not meaningfully associated with whether someone engages in a suicide attempt or behavior, but demonstrates a small, positive association with the number of past year attempts or behaviors. To our knowledge, the present study represents the most comprehensive evaluation of the association between aggression and suicide among veterans to date.

Our pattern of findings suggest that aggression may be a marker for frequent suicide attempts among high-risk veterans, aligning with civilian literature (Oquendo et al., 2021; Stanley et al., 2019). Such findings are important given that frequent suicidal behaviors are associated with lethality of suicidal behaviors, although the direction of this association is somewhat debated; some researchers suggests that a greater number of suicide attempts is negatively associated with attempt lethality (Kim et al., 2020), while others find that individuals who report multiple suicide attempts use more lethal means than individuals who report a single suicide attempt (Berardelli et al., 2020). Though our findings identify aggression as a marker of frequent suicidal behaviors, the association between aggression and suicidal behaviors was weak and should be interpreted with caution. Additionally, these findings apply to veteran at high-risk for suicide and may not generalize to all veterans.

4.1. Limitations

Regarding study limitations, we used cross-sectional data and were unable to speak to temporal precedence. Relatedly, the study was not designed to examine aggression as a causal mechanism of suicide and cannot speak to whether aggression causes suicidal behaviors. Also, our measure of aggression assessed trait aggression rather than the occurrence of discrete behaviors and does not capture the context in which aggression occurs. Additional research should attempt to replicate our findings using frequency measures of aggressive behaviors and when examining aggression across different contexts. Finally, as noted above, we recruited a sample of veterans at high-risk for suicide and our findings may not be generalizable to veterans across the continuum of suicide risk.

4.2. Implications and future directions

Future research should focus on understanding how and why aggression is related to the number of suicidal behaviors. For instance, further research should explore potential causal mechanisms of this association, including underlying mental health concerns (e.g., borderline personality disorder, posttraumatic stress disorder), common risk factors (e.g., combat exposure), or shared genetic vulnerability. Relatedly, our work suggests that findings related to aggression and suicidal behaviors are similar, yet not identical, when examining suicide attempts compared to suicidal behaviors. Future research should include a variety of suicidal behaviors when exploring predictors of suicide. Furthermore, there remains a pressing need for longitudinal research on the association between aggression and suicide among veterans (Schafer et al., 2022) to determine whether aggression predicts suicidal behaviors across time.

Clinically, our results suggest that veterans with high levels of aggression also report repeated suicide attempts or behaviors. The association between aggression and number of suicidal behaviors could inform strategies for identifying veterans at highest risk for suicidal behaviors. Relatedly, clinicians working with veterans at high-risk for suicide may consider assessing for aggression and referring to appropriate clinical services. Though the current findings are promising, they underscore the need for significantly more research on aggression and suicide among veterans to enhance suicide prevention efforts.

CRediT authorship contribution statement

Alison Krauss: Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Michael S. McCloskey:** Writing – review & editing, Conceptualization. **A. Solomon Kurz:** Writing – review & editing, Formal analysis. **Mark A. Ilgen:** Writing – review & editing. **Suzannah K. Creech:** Writing – review & editing. **Marianne Goodman:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Data curation.

Conflicts of interest

Dr. Marianne Goodman is a consultant for Boehringer Ingleheim Pharmacueticals.

The other authors have no conflicts of interest to disclose.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Marianne Goodman reports a relationship with Boehringer Ingelheim Pharmaceuticals Inc that includes: consulting or advisory. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Dr. Krauss is supported by the VISN 17 Center of Excellence for Research on Returning War Veterans, the Central Texas Veterans Health Care System, and Career Development Award IK1-CX002650 from the VA Clinical Science Research and Development Service (CSR&D). This research was supported by VA Merit Award I01-CX001705 awarded to Dr. Goodman from the VA CSR&D. Dr. Ilgen's time is supported by a Research Career Scientist Award from Health Services Research & Development (RCS 19–333). The views expressed herein are those of the authors and do not necessarily reflect the official policy or position of the Department of Veterans Affairs or the United States Government. The authors do not have any financial conflicts of interest to disclose.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpsychires.2025.06.001.

References

Berardelli, I., Forte, A., Innamorati, M., Imbastaro, B., Montalbani, B., Sarubbi, S., De Luca, G.P., Mastrangelo, M., Anibaldi, G., Rogante, E., Lester, D., Erbuto, D., Serafini, G., Amore, M., Pompili, M., 2020. Clinical differences between single and multiple suicide attempters, suicide ideators, and non-suicidal inpatients. Front. Psychiatr. 11, 605140. https://doi.org/10.3389/fpsyt.2020.605140.

Burke, T.A., Hamilton, J.L., Ammerman, B.A., Stange, J.P., Alloy, L.B., 2016. Suicide risk characteristics among aborted, interrupted, and actual suicide attempters. Psychiatry Res. 242, 357–364. https://doi.org/10.1016/j.psychres.2016.05.051.

Buss, A.H., Perry, M., 1992. The aggression Questionnaire. J. Pers. Soc. Psychol. 63, 452–459.

Chu, C., Wilks, C.R., Joiner, T., Gutierrez, P.M., 2023. Cross-sectional and longitudinal correlates of interrupted and aborted suicide attempts among U.S. active duty service members seeking treatment for suicidal symptoms. Clin. Psychol. Sci. 11, 863–878. https://doi.org/10.1177/21677026221150499.

Conner, K.R., Duberstein, P.R., Conwell, Y., Caine, E.D., 2003. Reactive aggression and suicide: theory and evidence. Aggress. Violent Behav. 8, 413–432. https://doi.org/10.1016/S1359-1789(02)00067-8.

Dewi, I.D.A.D.P., Kyranides, M.N., 2022. Physical, verbal, and relational aggression: the role of anger management strategies. J. Aggress. Maltreat. Trauma 31, 65–82. https://doi.org/10.1080/10926771.2021.1994495.

Dillon, K.H., Van Voorhees, E.E., Elbogen, E.B., 2020. Associations between anger and suicidal ideation and attempts: a prospective study using the National Epidemiologic Survey on Alcohol and Related Conditions. J. Psychiatr. Res. 122, 17–21. https://doi.org/10.1016/j.jpsychires.2019.12.011.

Elbogen, E.B., Johnson, S.C., Wagner, H.R., Newton, V.M., Timko, C., Vasterling, J.J., Beckham, J.C., 2012. Protective factors and risk modification of violence in Iraq and Afghanistan war veterans. J. Clin. Psychiatry 73, e767–e773. https://doi.org/10.4088/JCP.11m07593.

Enders, C.K., 2022. Applied Missing Data Analysis, second ed. The Guilford Press, New York.

Gvion, Y., 2018. Aggression, impulsivity, and their predictive value on medical lethality of suicide attempts: a follow-up study on hospitalized patients. J. Affect. Disord. 227, 840–846. https://doi.org/10.1016/j.jad.2017.11.033.

Gvion, Y., Apter, A., 2011. Aggression, impulsivity, and suicide behavior: a review of the literature. Arch. Suicide Res. 15, 93–112. https://doi.org/10.1080/13811118.2011.565265.

Gvion, Y., Levi-Belz, Y., 2018. Serious suicide attempts: systematic review of psychological risk factors. Front. Psychiatr. 9, 56. https://doi.org/10.3389/fpsyt.2018.00056.

Hillbrand, M., 2001. Threatening and non-threatening verbal aggression as predictors of physical aggression in violent psychiatric patients. J. Threat Assess. 1, 63–74. https://doi.org/10.1300/J177v01n02_04.

Kim, S., Choi, K.H., Lee, Kang-Sook, Kim, D., Hong, S., Lee, H., Kweon, Y., Lee, C.T., Lee, Kyoung-Uk, 2020. Risk factors for serious suicide attempts with high medical severity. Suicide Life. Threat. Beyond Behav. 50, 408–421. https://doi.org/10.1111/sltb.12597.

Mann, J.J., Arango, V.A., Avenevoli, S., Brent, D.A., Champagne, F.A., Clayton, P., Currier, D., Dougherty, D.M., Haghighi, F., Hodge, S.E., Kleinman, J., Lehner, T., McMahon, F., Mościcki, E.K., Oquendo, M.A., Pandey, G.N., Pearson, J., Stanley, B., Terwilliger, J., Wenzel, A., 2009. Candidate endophenotypes for genetic studies of suicidal behavior. Biol. Psychiatry 65, 556–563. https://doi.org/10.1016/j.biopsych.2008.11.021.

McCloskey, M.S., Ammerman, B.A., 2018. Suicidal behavior and aggression-related disorders. Curr. Opin. Psychol. 22, 54–58. https://doi.org/10.1016/j.copsyc.2017.08.010.

McGlade, E., Bueler, E., DiMuzio, J., Sheth, C., Legarreta, M., Yurgelun-Todd, D., 2021. Sex differences in suicidal behaviors and aggression in US Veterans. Psychiatry Res. 301, 113982. https://doi.org/10.1016/j.psychres.2021.113982.

Miller, I.W., Camargo, C.A., Arias, S.A., Sullivan, A.F., Allen, M.H., Goldstein, A.B., Manton, A.P., Espinola, J.A., Jones, R., Hasegawa, K., Boudreaux, E.D., for the ED-SAFE Investigators, 2017. Suicide prevention in an emergency department population: the ED-SAFE Study. JAMA Psychiatry 74, 563. https://doi.org/10.1001/jamapsychiatry.2017.0678.

Oquendo, M.A., Galfalvy, H.C., Choo, T.-H., Kandlur, R., Burke, A.K., Sublette, M.E., Miller, J.M., Mann, J.J., Stanley, B.H., 2021. Highly variable suicidal ideation: a phenotypic marker for stress induced suicide risk. Mol. Psychiatr. 26, 5079–5086. https://doi.org/10.1038/s41380-020-0819-0.

Posner, K., Brown, G.K., Stanley, B., Brent, D.A., Yershova, K.V., Oquendo, M.A., Currier, G.W., Melvin, G.A., Greenhill, L., Shen, S., Mann, J.J., 2011. The Columbia–Suicide Severity Rating Scale: initial validity and internal consistency findings from three multisite studies with adolescents and adults. Am. J. Psychiatr. 168, 1266–1277. https://doi.org/10.1176/appi.ajp.2011.10111704.

Schafer, K.M., Duffy, M., Kennedy, G., Stentz, L., Leon, J., Herrerias, G., Fulcher, S., Joiner, T.E., 2022. Suicidal ideation, suicide attempts, and suicide death among Veterans and service members: a comprehensive meta-analysis of risk factors. Mil. Psychol. 34, 129–146. https://doi.org/10.1080/08995605.2021.1976544.

Stanley, B., Michel, C.A., Galfalvy, H.C., Keilp, J.G., Rizk, M.M., Richardson-Vejlgaard, R., Oquendo, M.A., Mann, J.J., 2019. Suicidal subtypes, stress responsivity and impulsive aggression. Psychiatry Res. 280, 112486. https://doi.org/10.1016/j.psychres.2019.112486.

U.S. Department of Veterans Affairs, 2019. Department of Veterans Affairs FY 2018 - 2024 Strategic Plan.

U.S. Department of Veterans Affairs, 2023. Office of mental health and suicide prevention. 2023 national veteran suicide prevention annual report.

van Buuren, S., Groothuis-Oudshoorn, K., 2011. Mice : multivariate imputation by chained equations in R. J. Stat. Software 45. https://doi.org/10.18637/jss.v045.i03.

Varker, T., Cowlishaw, S., Baur, J., McFarlane, A.C., Lawrence-Wood, E., Metcalf, O., Van Hooff, M., Sadler, N., O'Donnell, M.L., Hodson, S., Benassi, H., Forbes, D., 2022. Problem anger in veterans and military personnel: prevalence, predictors, and associated harms of suicide and violence. J. Psychiatr. Res. 151, 57–64. https://doi.org/10.1016/j.jpsychires.2022.04.004.

Watkins, L.E., Sippel, L.M., Pietrzak, R.H., Hoff, R., Harpaz-Rotem, I., 2017. Co-occurring aggression and suicide attempt among veterans entering residential treatment for PTSD: the role of PTSD symptom clusters and alcohol misuse. J. Psychiatr. Res. 87, 8–14. https://doi.org/10.1016/j.jpsychires.2016.12.009.

Yau, K.K.W., Wang, K., Lee, A.H., 2003. Zero-inflated negative binomial mixed regression modeling of over-dispersed count data with extra zeros. Biom. J. 45, 437–452. https://doi.org/10.1002/bimj.200390024.